



# **JLOFC**

## PRODUCT PROFILE

HANGZHOU JINLONG OPTIC-ELECTRIC CO.,LTD





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# About Us

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Since  
1994



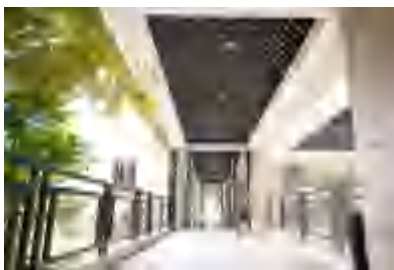
“The hometown of Wu Yue King , the hometown of cables”.Hangzhou Jinlong Optic-Electric Co., Ltd., founded in 1994, is located by the beautiful West Lake in Hangzhou. It mainly develops, produces and sells Optical Preforms, Fibers and Cables.

All performance and indicators of the optical fibers and cables exceed the requirements of international standards,and have obtained various product certificates and multiple international certificates such as CE,CPR. The products have been exported to over 30 countries and have gained high recognition from our partners.

In 2017, the company established its Wholly owned subsidiary - HANGZHOU JINXINGTONG OPTICAL FIBER TECHNOLOGY CO., LTD.,becoming a professional enterprise dedicated to the entire industrial chain of optical preform ,optical fiber, and optical cable . It has completed an annual production capacity layout of 300 tons of preforms, 12 million fiber kilometers of fibers, and 10 million fiber kilometers of cables.

The company has advanced technology and has a top-notch R&D and technical service team for preforms, fibers and cables in the industry. It has successively won the titles of National High tech Enterprise and National Specialized and Innovative "Little Giant" Enterprise. And we have carried out technical cooperation with top domestic universities and research institutes, and our company's technology center has been rated as a key laboratory in Zhejiang Province. The independently developed technologies such as core deposition, core sintering, clad deposition, clad sintering, fiber drawing, and cable manufacturing have all reached the world's leading level.Optical fiber, as a communication transmission medium, provides a solid transmission foundation for the development of 5G networks. Meanwhile, the widespread application of fiber in AI life is profoundly changing our future. With the continuous advancement of technology and the expansion of applications, the combination of fiber, 5G, and AI technology will bring more surprises to our future.

Win-win cooperation , achieving dreams.Paving infinite possibilities for tomorrow's information superhighway.



The company has obtained certifications including ISO9001 Quality Management System, ISO14001 Environmental Management System, ISO45001 Occupational Health and Safety Management System, ISO10012 Measurement Management System, and TL9000. It has also been awarded titles such as "High-Tech Enterprise", "AAA-level Integrity Management Demonstration Unit", "AAA-level Credit Enterprise", and "Enterprise with Social Responsibility Construction".

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In 2017, the company established a holding subsidiary - Hangzhou Jinxingtong Optical Fiber Technology Co., Ltd., becoming an enterprise specializing in the entire industrial chain of optical rod, optical fiber and optical cable communication. Meanwhile, it was listed as a key project in Zhejiang Province's "152" Project.

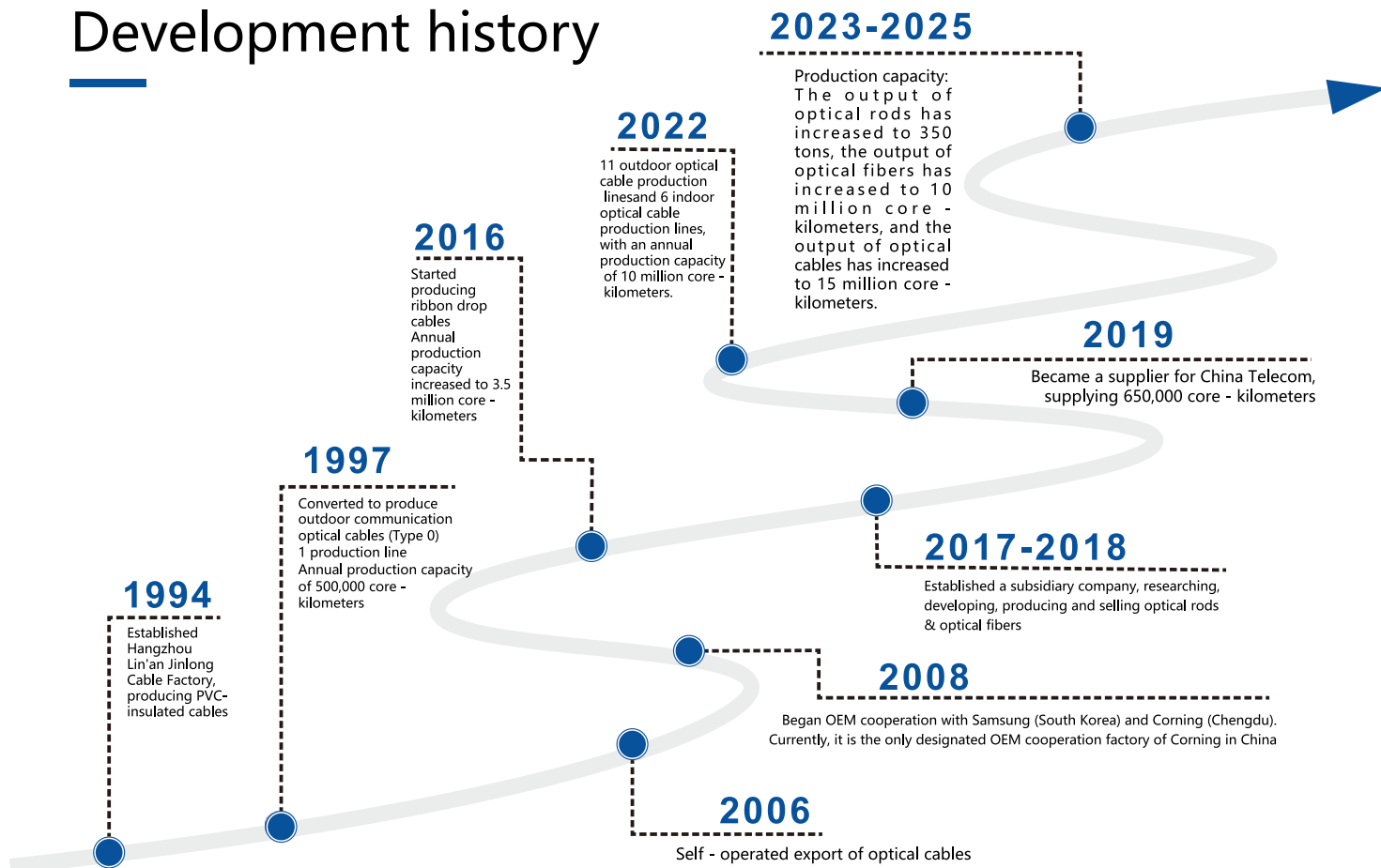
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**3M**   **10M**   **12M**

At present, the company has achieved an annual production capacity of 300 tons of fiber optic preforms, 10 million fiber-kilometers of optical fiber, and 12 million fiber-kilometers of optical cables.

# COMPANY DEVELOPMENT

## Development history



# CERTIFICATIONS

## Qualification Certificates



# PRODUCTS

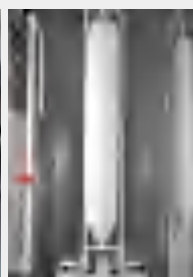
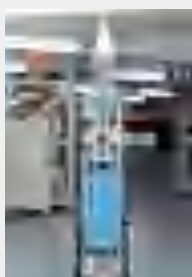
## Introduction to Optical Rod Products

### Optical Fiber Preform

**Product Description** High-purity synthetic quartz prepared by the flame hydrolysis method.

#### Product Features

It has a series of optical fiber preforms that meet the requirements for drawing G.652D, G657A1, G657A2 and special optical fibers. The indicators of optical fiber products meet and exceed the ITU-T standards.



### Non-dispersion-shifted single-mode optical fiber with extended wavelength range

**Product Description** Meets and exceeds the requirements of ITU-T G.652D & IEC B1.3 optical fiber technical specifications.

#### Product Features

The optical fiber eliminates the 1383nm water peak of non-dispersion-shifted fiber, expanding the operating wavelength range to 1360 - 1625nm, and is suitable for 1360 - 1625nm metropolitan area network CWDM transmission.

G652D



## Bend-insensitive single-mode optical fiber (G.657A1)

G.657A1

**Product Description** Meets and exceeds the requirements of ITU-T G.657.A1 optical fiber technical specifications.

### Product Features

It meets all the transmission characteristics of G.652 optical fiber, with extremely low microbend additional attenuation; it has a high fatigue resistance parameter, ensuring service life under small bending radii.



A1BB

**Product Description** Meets and exceeds the requirements of ITU-T G.652.D/ G.657A1 optical fiber technical specifications.

### Product Features

Large mode field diameter, perfectly compatible with G.652 optical fiber, and extremely low microbend additional attenuation; It has a high fatigue resistance parameter, ensuring service life under small bending radii.

G.657A2

**Product Description** Meets and exceeds the requirements of ITU-T G.657.A1/A2 optical fiber technical specifications.

### Product Features

The minimum bending radius can reach 7.5mm, with excellent bending resistance performance.

A2BB

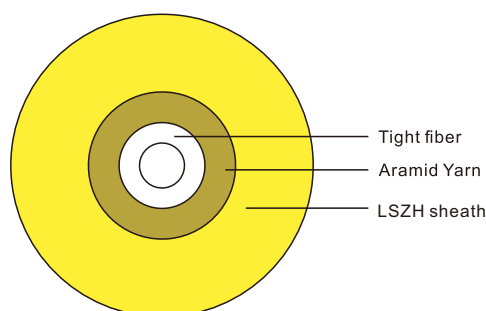
**Product Description** Meets and exceeds the requirements of ITU-T G.652.D/ G.657A2 optical fiber technical specifications.

### Product Features

Large mode field diameter ( $\geq 8.8\mu\text{m}$ ), perfectly compatible with G.652 optical fiber, and extremely low microbend additional attenuation; It has a high fatigue resistance parameter, ensuring service life under small bending radii.

# SXC

## Single-core indoor optical cable



### Product Description

Single-core indoor optical cable uses 900 $\mu$ m or 600 $\mu$ m tight-buffered optical fiber as the optical transmission unit, covered with aramid yarn as the strength member, and extruded with a layer of polyvinyl chloride (PVC) sheath on the outermost layer. Sheaths made of materials such as low smoke zero halogen (LSZH) can also be provided according to customer requirements.

### Product Features

- Adopts tight-buffered optical fiber, which is easy to strip;
- The tight-buffered optical fiber has good flame retardant performance;
- Aramid strength members endow the optical cable with excellent tensile performance;
- The outer sheath material has the advantages of corrosion resistance, water resistance, flame retardancy, environmental protection, etc.

### Product Applications

- Optical fiber active connection jumpers or pigtails
- Indoor riser-level and forced air ventilation-level cabling
- Interconnection of instruments and communication equipment

### Structural Parameters

| Cable Model | Reference Cable Diameter (mm) | Reference Cable Weight (kg/km) |             | Tight-Buffered Fiber Diameter ( $\mu$ m) | Allowable Tensile Force Long-term/Short-term (N) | Allowable Crush Force Long-term/Short-term (N) | Bending Radius Static/Dynamic (mm) |
|-------------|-------------------------------|--------------------------------|-------------|------------------------------------------|--------------------------------------------------|------------------------------------------------|------------------------------------|
|             |                               | PVC Sheath                     | LSZH Sheath |                                          |                                                  |                                                |                                    |
| SXC-1.6     | 1.6 $\pm$ 0.2                 | 2.4                            | 2.6         | 600 $\pm$ 50                             | 40/80                                            | 100/500                                        | 60/30                              |
| SXC-1.9     | 1.9 $\pm$ 0.2                 | 3.5                            | 4.5         | 900 $\pm$ 50                             | 60/100                                           | 100/500                                        | 60/30                              |
| SXC-2.8     | 2.8 $\pm$ 0.2                 | 6.0                            | 7.5         | 900 $\pm$ 50                             | 60/100                                           | 100/500                                        | 60/30                              |

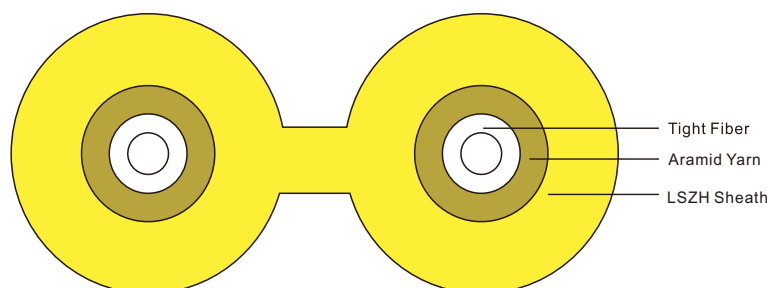
### Optical Characteristics

| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
| Conditions         | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
|                    | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 单位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G652D              | G652D         | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 16                     |
| G657A1             | G657A1        | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| 50/125             | 50/125        | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 500/500              | ---                       | ---                      | 30                     |
| 62.5/125           | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 200/500              | ---                       | ---                      | 30                     |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 1500/500             | $\geq$ 2000               | $\leq$ 300               | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 3500/500             | $\geq$ 4700               | $\leq$ 550               | 30                     |
| BI-OM3             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 1500/500             | $\geq$ 2000               | $\leq$ 300               | 7.5                    |
| BI-OM4             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 3500/500             | $\geq$ 4700               | $\leq$ 550               | 7.5                    |

Transportation/storage/operating temperature: -20 $^{\circ}$ C ~ +60 $^{\circ}$ C, installation temperature: -5 $^{\circ}$ C ~ +50 $^{\circ}$ C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# Dual-core figure-8 optical cable



## Product Description

The 2-core figure-8 optical cable uses 900 $\mu$ m or 600 $\mu$ m tight-buffered optical fibers as the optical transmission unit, covered with aramid yarn as the strength member, and extruded with a figure-8 polyvinyl chloride (PVC) sheath on the outermost layer. Sheaths made of low smoke zero halogen (LSZH) material can also be provided according to customer requirements.

## Product Features

- Adopts tight-buffered optical fiber for easy stripping;
- The tight-buffered optical fiber has good flame retardancy;
- Aramid strength members endow the optical cable with excellent tensile performance;
- The outer sheath material has the advantages of corrosion resistance, water resistance, flame retardancy and environmental protection;
- The figure-8 structure is easy to strip and use in branches.

## Product Applications

- Duplex optical fiber active connection jumpers or pigtails
- Indoor riser-level and forced air ventilation-level cabling
- Interconnection of instruments and communication equipment

## Structural Parameters

| Cable Model | Reference Cable Diameter (mm)  | Reference Cable Weight (kg/km) |             | Tight-Buffered Fiber Diameter ( $\mu$ m) | Allowable Tensile Force Long-term/Short-term (N) | Allowable Crush Force Long-term/Short-term (N) | Bending Radius Static/Dynamic (mm) |
|-------------|--------------------------------|--------------------------------|-------------|------------------------------------------|--------------------------------------------------|------------------------------------------------|------------------------------------|
|             |                                | PVC Sheath                     | LSZH Sheath |                                          |                                                  |                                                |                                    |
| SXC-1.6     | $(3.4\pm0.3)\times(1.6\pm0.3)$ | 4.8                            | 5.3         | 600 $\pm$ 50                             | 40/80                                            | 60/30                                          | 60/30                              |
| SXC-1.9     | $(4.0\pm0.3)\times(1.6\pm0.3)$ | 8.0                            | 8.7         | 900 $\pm$ 50                             | 100/200                                          | 60/30                                          | 60/30                              |
| SXC-2.8     | $(5.8\pm0.3)\times(1.6\pm0.3)$ | 11.6                           | 14.8        | 900 $\pm$ 50                             | 100/200                                          | 60/30                                          | 60/30                              |

## Optical Characteristics

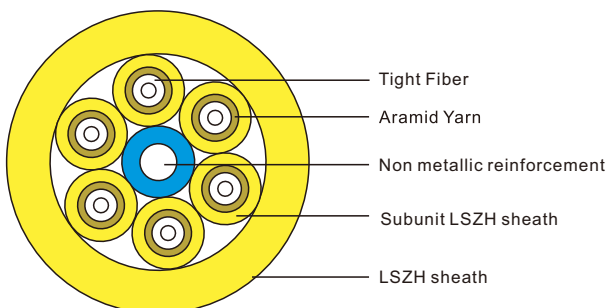
| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
|                    | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
| Conditions         | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 单位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G652D              | G652D         | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 16                     |
| G657A1             | G657A1        | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| 50/125             | 50/125        | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 500/500              | ---                       | ---                      | 30                     |
| 62.5/125           | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 200/500              | ---                       | ---                      | 30                     |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 1500/500             | $\geq$ 2000               | $\leq$ 300               | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 3500/500             | $\geq$ 4700               | $\leq$ 550               | 30                     |
| BI-OM3             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 1500/500             | $\geq$ 2000               | $\leq$ 300               | 7.5                    |
| BI-OM4             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq$ 3500/500             | $\geq$ 4700               | $\leq$ 550               | 7.5                    |

Transportation/storage/operating temperature: -20°C ~ +60°C, installation temperature: -5°C ~ +50°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

## BOC

## Multi-Purpose Break-out Cable



## Product Description

Will adjustments to product dimensions and performance values affect product quality?

The multi-purpose branch optical cable uses single-core optical cables (900μm tight-buffered optical fibers with aramid strength members) as sub-units, with a non-metallic central strength member. The cable sub-units are stranded around the central strength member to form the cable core. The outermost layer is extruded with a polyvinyl chloride (PVC) sheath. Sheaths made of low smoke zero halogen (LSZH) material can also be provided according to customer requirements.

## Product Features

The non-metallic strength member enables the optical cable to withstand greater tensile force.

- ♦ Aramid strength members endow the optical cable with excellent tensile performance.
- ♦ The outer sheath material has the advantages of corrosion resistance, water resistance, flame retardancy and environmental protection.

## Product Applications

- Indoor generic cabling
- Building backbone cabling optical cable

## ● Structural Parameters

| Optical Cable Model | Reference Cable Diameter (mm) | Reference Cable Weight (kg/km) |     | Allowable Tensile Force (Long-term/Short-term) | Allowable Crush Force (Long-term/Short-term) | Bending Radius (Static/Dynamic) |
|---------------------|-------------------------------|--------------------------------|-----|------------------------------------------------|----------------------------------------------|---------------------------------|
| BOC-X-XX-04Xn       | 7.4±0.5                       | 50                             | 56  | 130/440                                        | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-06Xn       | 8.4±0.5                       | 66                             | 74  | 200/660                                        | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-08Xn       | 9.8±0.5                       | 91                             | 101 | 200/660                                        | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-12Xn       | 12.4±0.5                      | 140                            | 155 | 200/660                                        | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-18Xn       | 12.4±0.5                      | 140                            | 155 | 400/1320                                       | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-24Xn       | 14.4±0.5                      | 181                            | 201 | 400/1320                                       | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-36Xn       | 16.4±0.5                      | 208                            | 255 | 400/1320                                       | 300/1000                                     | 20D/10D                         |
| BOC-X-XX-48Xn       | 19±0.5                        | 280                            | 344 | 400/1320                                       | 300/1000                                     | 20D/10D                         |

## ● Optical Characteristics

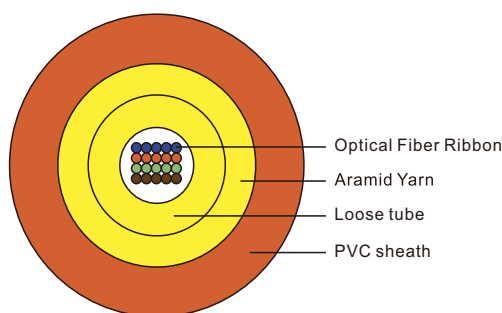
| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
| Conditions         | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
|                    | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 单位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G652D              | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 16                     |
| G657A1             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| 50/125             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥500/500                    | ---                       | ---                      | 30                     |
| 62.5/125           | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥200/500                    | ---                       | ---                      | 30                     |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 30                     |
| BI-OM3             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 7.5                    |
| BI-OM4             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 7.5                    |

Transportation/Storage/Operation Temperature: -20°C ~ +60°C; Installation Temperature: -5°C ~ +50°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# GJFDV

## Indoor fiber ribbon cable



### Product Description

The dual-core figure-8 optical cable uses 900 $\mu$ m or 600 $\mu$ m tight-buffered optical fibers as the optical transmission unit, covered with aramid yarn as the strength member, and the outermost layer is extruded with a figure-8 polyvinyl chloride (PVC) sheath. Sheaths made of low smoke zero halogen (LSZH) material can also be provided according to customer requirements.

### Product Features

- Tight-buffered optical fibers are adopted for easy stripping;
  - Tight-buffered optical fibers have good flame retardant performance;
  - Aramid strength members endow the optical cable with excellent tensile performance;
  - The outer sheath material has the advantages of corrosion resistance, water resistance, flame retardancy and environmental protection;
- The figure-8 structure is easy to peel off for branch use.

### Product Applications

- Duplex fiber optic patch cords or pigtails
- Indoor riser and forced air ventilation cabling
- Interconnection of instruments and communication equipment

### Structural Parameters

| Optical Cable Model | Reference Cable Diameter (mm) | Reference Cable Weight (kg/km) | Allowable Tensile Force (Long-term/Short-term) | Allowable Crush Force (Long-term/Short-term) | Bending Radius (Static/Dynamic) |
|---------------------|-------------------------------|--------------------------------|------------------------------------------------|----------------------------------------------|---------------------------------|
| GJFDV-X-48Xn        | 12Fx4                         | 84                             | 200/660                                        | 300/1000                                     | 25D/12.5D                       |
| GJFDV-X-72Xn        | 12Fx6                         | 90                             | 200/660                                        | 300/1000                                     | 25D/12.5D                       |
| GJFDV-X-96Xn        | 12Fx8                         | 100                            | 200/660                                        | 300/1000                                     | 25D/12.5D                       |
| GJFDV-X-144Xn       | 12Fx12                        | 125                            | 200/660                                        | 300/1000                                     | 25D/12.5D                       |

### Optical Characteristics

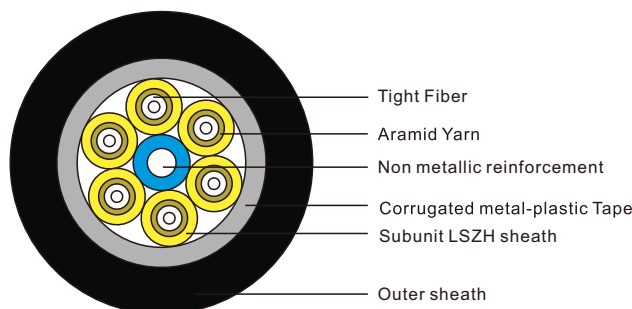
| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
| Conditions         | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
|                    | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 単位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G657A1             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| 50/125             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥500/500                    | ---                       | ---                      | 30                     |
| 62.5/125           | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥200/500                    | ---                       | ---                      | 30                     |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 30                     |
| BI-OM3             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 7.5                    |
| BI-OM4             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 7.5                    |

Transportation/Storage/Service Temperature: -20°C ~ +60°C; Installation Temperature: -5°C ~ +50°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

## GJA

## Waterproof Optical Fiber Pigtail



## Product Description

The waterproof pigtail uses a single - core optical cable (900μm tight - buffered fiber, aramid strength member) as a sub - unit, with a non - metallic central strength member. The optical cable sub - units are stranded around the central strength member to form the cable core. An aluminum tape moisture - proof layer is wrapped outside, and a polyethylene sheath is extruded as the outermost layer.

## Product Features

- The non-metallic strength member enables the optical cable to withstand greater tensile force.
- The aluminum tape has good moisture - proof performance.
- The outer sheath material has the advantages of corrosion resistance, water resistance, flame retardancy, and environmental protection.

## Product Applications

- Indoor structured cabling
- Connection of outdoor-to-indoor optical cable distribution equipment

## ● Structural Parameters

| Optical Cable Model | Reference Cable Diameter (mm) | Reference Cable Weight (kg/km) |     | Allowable Tensile Force (Long-term/Short-term) | Allowable Crush Force (Long-term/Short-term) | Bending Radius (Static/Dynamic) |
|---------------------|-------------------------------|--------------------------------|-----|------------------------------------------------|----------------------------------------------|---------------------------------|
| GJA-II-YX-4Xn       | 8.3±0.5                       | 58                             | 58  | 130/440                                        | 300/1000                                     | 20D/10D                         |
| GJA-II-YX-6Xn       | 9.2±0.5                       | 73                             | 73  | 200/660                                        | 300/1000                                     | 20D/10D                         |
| GJA-II-YX-8Xn       | 10.6±0.5                      | 96                             | 96  | 200/660                                        | 300/1000                                     | 20D/10D                         |
| GJA-II-YX-12Xn      | 13.4±0.5                      | 115                            | 155 | 200/660                                        | 300/1000                                     | 20D/10D                         |

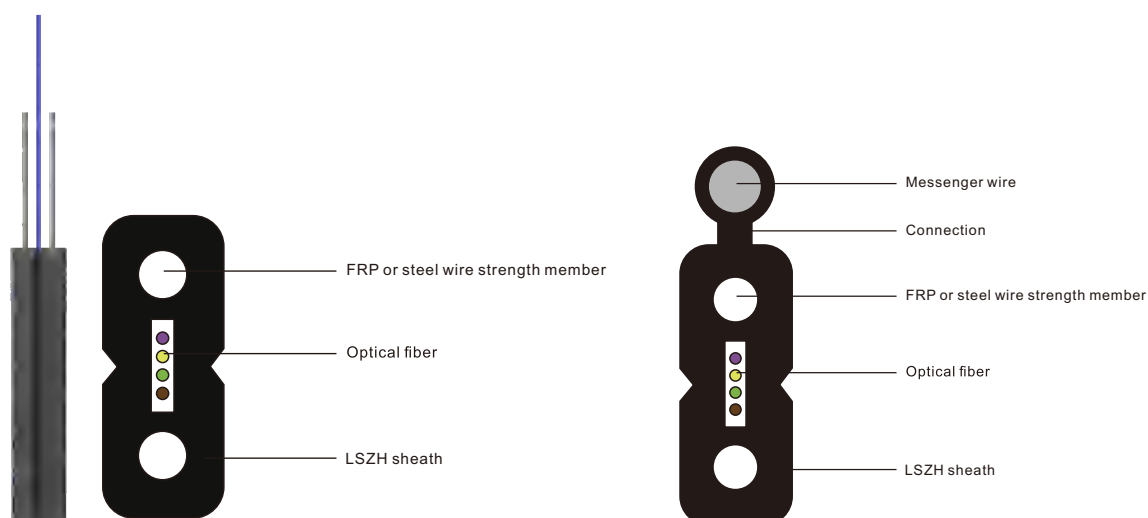
## ● Optical Characteristics

| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
| Conditions         | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
|                    | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 单位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G652D              | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 16                     |
| G657A1             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| 50/125             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥500/500                    | ---                       | ---                      | 30                     |
| 62.5/125           | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥200/500                    | ---                       | ---                      | 30                     |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 30                     |
| BI-OM3             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥1500/500                   | ≥2000                     | ≤300                     | 7.5                    |
| BI-OM4             | ---           | ---           | 3.0/1.0       | 3.5/1.5       | ≥3500/500                   | ≥4700                     | ≤550                     | 7.5                    |

Transportation/Storage/Operating Temperature: -20°C ~ +60°C; Installation Temperature: -5°C ~ +50°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# Indoor Bow - type Drop Cables for Access Network



## Product Description

The colored fiber is placed into a loose tube made of high-modulus material, and the loose tube is filled with thixotropic water-blocking fiber paste. At the center of the cable core is a metal strength member. The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core. The gaps of the cable core and between the cable core and the steel tape are filled with water-blocking cable paste. A polyethylene inner sheath is extruded over the cable core, and after the double-sided coated steel tape is longitudinally lapped, a polyethylene sheath is extruded to form the cable.

## Product Features

- Special bend-resistant optical fibers provide greater bandwidth and enhance network transmission characteristics.
- Two parallel FRPs endow the optical cable with good compressive performance to protect the optical fibers.
- The optical cable features a simple structure, light weight and strong practicability.
- It is easy to strip and splice, simplifying installation and maintenance.
- Low smoke zero halogen (LSZH) flame-retardant sheath, environmentally friendly.

## Structural Parameters

| Optical Cable Model | Reference Cable Diameter (mm) | Reference Cable Weight (kg/km)       | Allowable Tensile Force (Long-term/Short-term) | Allowable Crush Force (Long-term/Short-term) | Bending Radius (Static/Dynamic) | Optical Cable Model |
|---------------------|-------------------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|---------------------------------|---------------------|
| GJXFH-1Xn           | 1                             | $(2.0 \pm 0.1) \times (3.0 \pm 0.1)$ | 8                                              | 40/80                                        | 500/1000                        | 15/30               |
| GJXFH-2Xn           | 2                             | $(2.0 \pm 0.1) \times (3.0 \pm 0.1)$ | 8.5                                            | 40/80                                        | 500/1000                        | 15/30               |
| GJXFH-1Xn           | 1                             | $(2.0 \pm 0.1) \times (3.8 \pm 0.1)$ | 9                                              | 40/80                                        | 500/1000                        | 15/30               |
| GJXFH-4Xn           | 4                             | $(2.0 \pm 0.2) \times (4.0 \pm 0.2)$ | 10                                             | 40/80                                        | 500/1000                        | 15/30               |

## Optical Characteristics

| Optical Fiber Type | Attenuation   |               |               |               | Overfilled Launch Bandwidth | Effective Modal Bandwidth | 10G Ethernet Link Length | Minimum Bending Radius |
|--------------------|---------------|---------------|---------------|---------------|-----------------------------|---------------------------|--------------------------|------------------------|
| Conditions         | 1310/1550nm   |               | 1310/1550nm   |               | 850/1300nm                  | 850nm                     | 850nm                    | /                      |
|                    | Typical Value | Maximum Value | Typical Value | Maximum Value |                             |                           |                          |                        |
| 单位                 | dB/km         | dB/km         | dB/km         | dB/km         | MHZ.km                      | MHZ.km                    | m                        | mm                     |
| G657A1             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 10                     |
| G657A2             | 0.36/0.22     | 0.5/0.4       | ---           | ---           | ---                         | ---                       | ---                      | 7.5                    |
| OM3                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq 500/500$              | $\geq 2000$               | $\leq 300$               | 30                     |
| OM4                | ---           | ---           | 3.0/1.0       | 3.5/1.5       | $\geq 200/500$              | $\geq 4700$               | $\leq 550$               | 30                     |

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# (GYTS/GYTA)

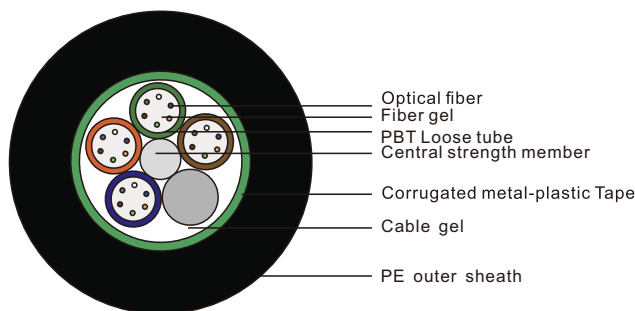
## Loose tube stranded light armored optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste.

At the center of the cable core is a metal strength member. The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core. The gaps in the cable core and the space between the cable core and the steel tape are filled with water-blocking cable paste. After the double-sided coated steel tape is longitudinally wrapped, a polyethylene sheath is extruded.



### Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- The tube is filled with special grease, which provides key sealing protection for the optical fibers.
- The PE sheath has excellent UV radiation resistance.
- The single steel wire central strength member helps with the alignment and tensile resistance of the optical cable.
- It has excellent mechanical properties and temperature characteristics.
- The double-sided corrugated steel tape (PSP) enhances the cable's moisture resistance, and the corrugated part combines better with PE, making the structure more robust.

### Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### Optical Characteristics

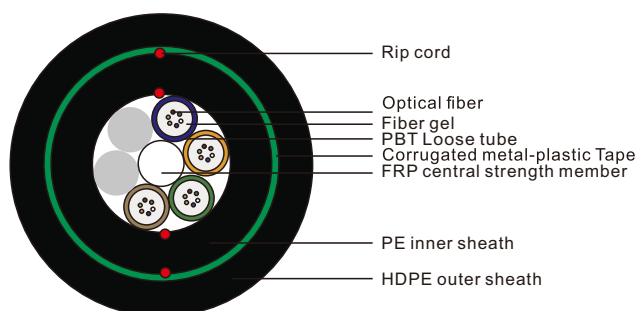
| Optical Cable Model<br>(Increasing by 2 Fibers) | Number of Fibers | Number of Sleeves | Number of Filler Ropes | Reference Optical Cable Weight<br>(kg/km) | Allowable Tensile Force (Long-term /Short-term) (N) | Allowable Crush Force (Long-term/Short-term) (N) | Bending Radius (Static/Dynamic) (mm) |
|-------------------------------------------------|------------------|-------------------|------------------------|-------------------------------------------|-----------------------------------------------------|--------------------------------------------------|--------------------------------------|
| GYTS-24Xn                                       | 24               | 4                 | 1                      | 96                                        | 600/1500                                            | 300/1000                                         | 10D/20D                              |
| GYTS-48Xn                                       | 48               | 4                 | 1                      | 111                                       |                                                     |                                                  |                                      |
| GYTS-96Xn                                       | 96               | 8                 | 0                      | 168                                       |                                                     |                                                  |                                      |
| GYTS-144Xn                                      | 144              | 12                | 0                      | 228                                       |                                                     |                                                  |                                      |
| GYTS-288Xn                                      | 288              | 24                | 0                      | 283                                       |                                                     |                                                  |                                      |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

# (GY(F)TY53/GY(F)TA53)

## Loose Tube Stranded Heavy - duty Armored Optical Cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste.

At the center of the cable core is a metal strength member. The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core. The gaps in the cable core and the space between the cable core and the aluminum tape are filled with water-blocking cable paste. After the plastic-coated aluminum tape is longitudinally wrapped, a polyethylene sheath is extruded.

### Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- The tube is filled with special grease, which provides key sealing protection for the optical fibers.
- The PE sheath has excellent UV radiation resistance.
- The single steel wire central strength member helps with the alignment and tensile resistance of the optical cable.
- It has excellent mechanical properties and temperature characteristics.
- The double-sided corrugated steel tape (PSP) enhances the cable's moisture resistance, and the corrugated part combines better with PE, making the structure more robust.
- High-quality water-blocking materials prevent longitudinal water seepage of the optical cable.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

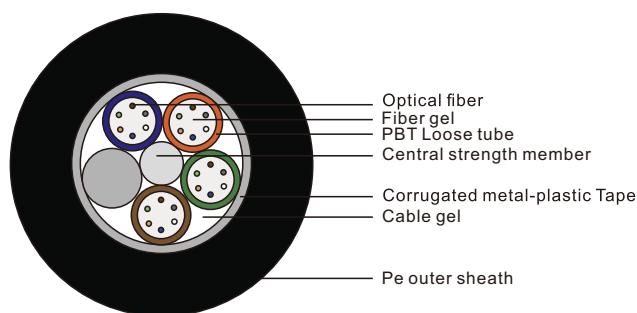
### ● Optical Characteristics

| Optical Cable Model<br>(Increasing by 2 Fibers) | Number of Fibers | Number of Sleeves | Number of Filler Ropes | Reference Optical Cable Weight<br>(kg/km) | Allowable Tensile Force (Long-term/Short-term) (N) | Allowable Crush Force (Long-term/Short-term) (N) | Bending Radius (Static/Dynamic) (mm) |
|-------------------------------------------------|------------------|-------------------|------------------------|-------------------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------|
| GYTY53-24Xn                                     | 24               | 4                 | 1                      | 160                                       | 1000/3000                                          | 1000/3000                                        | 12.5D/25D                            |
| GYTY53-48Xn                                     | 48               | 4                 | 1                      | 174                                       |                                                    |                                                  |                                      |
| GYTY53-96Xn                                     | 96               | 8                 | 0                      | 214                                       |                                                    |                                                  |                                      |
| GYTY53-144Xn                                    | 144              | 12                | 0                      | 281                                       |                                                    |                                                  |                                      |

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

# (GYFTS/GYFTA)

## Loose tube stranded armored optical cable with non-metallic strength member



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste. At the center of the cable core is a glass fiber reinforced plastic (FRP). The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core, and the gaps in the cable core are filled with water-blocking cable paste. After the plastic-coated aluminum tape is longitudinally wrapped, a polyethylene sheath is extruded to form the cable.

### Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- It has excellent mechanical properties and temperature characteristics.
- The tube is filled with special grease, which provides key sealing protection for the optical fibers.
- The PE sheath has excellent UV radiation resistance.
- A single non-metallic central strength member with plastic-coated aluminum tape moisture-proof layer.

### Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### Optical Characteristics

| Optical Cable Model<br>(Increasing by 2 Fibers)) | Number of Fibers | Number of Sleeves | Number of Filler Ropes | Reference Optical<br>Cable Weight<br>(kg/km) | Allowable Tensile<br>Force (Long-term<br>/Short-term) (N) | Allowable Crush<br>Force (Long-<br>term/Short-term)<br>(N) | Bending Radius<br>(Static/Dynamic)<br>(mm) |
|--------------------------------------------------|------------------|-------------------|------------------------|----------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|--------------------------------------------|
| GYFTA-24Xn                                       | 24               | 4                 | 2                      | 99                                           | 400/1000                                                  | 300/1000                                                   | 10D/20D                                    |
| GYFTA-24Xn                                       | 24               | 4                 | 3                      | 111                                          | 600/1500                                                  |                                                            |                                            |
| GYFTA-48Xn                                       | 48               | 4                 | 2                      | 122                                          |                                                           |                                                            |                                            |
| GYFTA-24Xn                                       | 24               | 4                 | 4                      | 127                                          | 1000/3000                                                 |                                                            |                                            |
| GYFTA-48Xn                                       | 48               | 8                 | 0                      | 127                                          |                                                           |                                                            |                                            |
| GYFTA-96Xn                                       | 96               | 8                 | 0                      | 136                                          |                                                           |                                                            |                                            |
| GYFTA-144Xn                                      | 144              | 12                | 0                      | 195                                          |                                                           |                                                            |                                            |

Storage and service temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

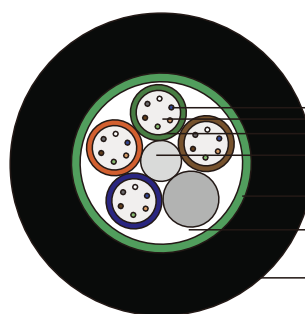
# (GYTZS/GYTZA)

## Loose tube stranded flame-retardant optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste. At the center of the cable core is a metal strength member. The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core, and the gaps in the cable core are filled with water-blocking cable paste. After the double-sided coated steel tape is longitudinally wrapped, a flame-retardant sheath is extruded to form the cable.



Optical fiber  
Fiber gel  
PBT Loose tube  
Central strength member  
Corrugated metal-plastic Tape  
Cable gel  
PE outer sheath

### Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- ♦ The tube is filled with special grease, which provides key sealing protection for the optical fibers.
- ♦ The special sheath has excellent flame retardancy.
- ♦ The single steel wire central strength member helps with the alignment and tensile resistance of the optical cable.
- ♦ It has excellent mechanical properties and temperature characteristics.
- ♦ The double-sided corrugated steel tape (PSP) enhances the cable's moisture resistance.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

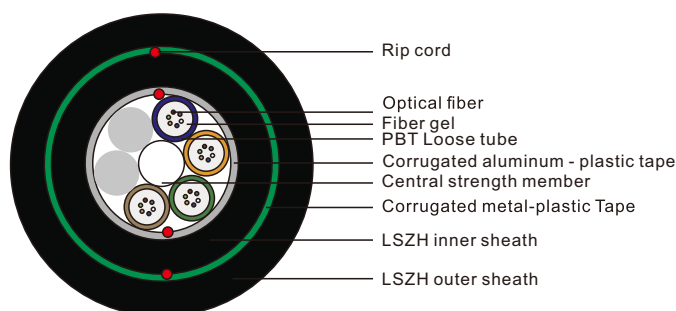
### ● Optical Characteristics

| Optical Cable Model<br>(Increasing by 2 Fibers) | Number of Fibers | Number of Sleeves | Number of Filler Ropes | Reference Optical Cable Weight<br>(kg/km) | Allowable Tensile Force (Long-term /Short-term) (N) | Allowable Crush Force (Long-term/Short-term) (N) | Bending Radius (Static/Dynamic) (mm) |
|-------------------------------------------------|------------------|-------------------|------------------------|-------------------------------------------|-----------------------------------------------------|--------------------------------------------------|--------------------------------------|
| GYTZS-24Xn                                      | 24               | 4                 | 1                      | 118                                       | 600/1500                                            | 300/1000                                         | 10D/20D                              |
| GYTZS-48Xn                                      | 48               | 4                 | 1                      | 139                                       |                                                     |                                                  |                                      |
| GYTZS-96Xn                                      | 96               | 8                 | 0                      | 200                                       |                                                     |                                                  |                                      |
| GYTZS-144Xn                                     | 144              | 12                | 0                      | 268                                       |                                                     |                                                  |                                      |
| GYTZS-288Xn                                     | 288              | 24                | 0                      | 360                                       |                                                     |                                                  |                                      |

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

# (GY(F)TZA53)

Loose tube stranded armored flame-retardant optical cable with strength member



## Product Description

Colored optical fibers are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste.

At the center of the cable core is a non-metallic strength member (FRP). The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core. The gaps in the cable core and the space between the cable core and the steel tape are filled with water-blocking cable paste. After the plastic-coated aluminum tape is longitudinally wrapped, a polyethylene inner sheath is extruded; then, after the double-sided coated steel tape is longitudinally wrapped, a flame-retardant outer sheath is extruded to form the cable.

## Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- ♦ The tube is filled with special grease, providing key sealing protection for the optical fibers.
- ♦ The special sheath has excellent flame retardancy.
- ♦ A single non-metallic central strength member.
- ♦ It has excellent mechanical properties and temperature characteristics.
- ♦ Plastic-coated aluminum tape moisture-proof layer.
- ♦ The double-sided corrugated steel tape (PSP) enhances the cable's moisture resistance.

## Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

## Optical Characteristics

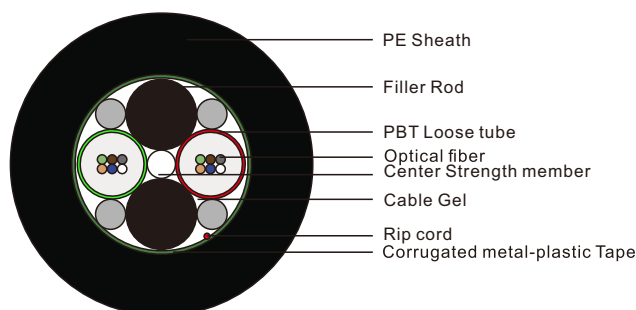
| Optical Cable Model<br>(Increasing by 2 Fibers)) | Number of Fibers | Number of Sleeves | Number of Filler Ropes | Reference Optical Cable Weight<br>(kg/km) | Allowable Tensile Force (Long-term /Short-term) (N) | Allowable Crush Force (Long-term/Short-term) (N) | Bending Radius<br>(Static/Dynamic)<br>(mm) |
|--------------------------------------------------|------------------|-------------------|------------------------|-------------------------------------------|-----------------------------------------------------|--------------------------------------------------|--------------------------------------------|
| GYTS-24Xn                                        | 24               | 4                 | 1                      | 96                                        | 600/1500                                            | 300/1000                                         | 10D/20D                                    |
| GYTS-48Xn                                        | 48               | 4                 | 1                      | 111                                       |                                                     |                                                  |                                            |
| GYTS-96Xn                                        | 96               | 8                 | 0                      | 168                                       |                                                     |                                                  |                                            |
| GYTS-144Xn                                       | 144              | 12                | 0                      | 228                                       |                                                     |                                                  |                                            |
| GYTS-288Xn                                       | 288              | 24                | 0                      | 283                                       |                                                     |                                                  |                                            |

Storage and service temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# (GYDTS/GYDTA)

## Loose tube stranded light armored fiber ribbon cable



### Product Description

4, 6, 8, and 12-fiber ribbons are placed into loose tubes made of high-modulus materials, and the loose tubes are filled with thixotropic water-blocking fiber paste.

At the center of the cable core is a metal strength member. The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core, with the gaps inside the cable core filled with water-blocking cable paste. After the plastic-coated aluminum tape is longitudinally wrapped, a polyethylene sheath is extruded to form the cable.

### Product Features

- The loose tube material itself has good hydrolysis resistance and high strength.
- The tube is filled with special grease, providing key sealing protection for the optical fibers.
- The special sheath has excellent flame retardancy.
- A single non-metallic central strength member.
- 2, 6, 8, and 12-fiber ribbons are adopted, allowing flexible application.
- It has excellent mechanical properties and temperature characteristics.
- Plastic-coated aluminum tape moisture-proof layer.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### ● Optical Characteristics

| Optical Cable Model<br>(Increasing by 2<br>Fibers) | Number of<br>Fibers | Number of<br>sleeves | Number of<br>filler ropes | Reference Optical Cable<br>Weight<br>(kg/km) | Allowable<br>tensile force Long-<br>term / Short-term<br>(N) | Allowable<br>Crush Force (Long-<br>term/Short-term)<br>(N) | Bending Radius<br>(Static/Dynamic)<br>(mm) |
|----------------------------------------------------|---------------------|----------------------|---------------------------|----------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------|
| GYDTA48Xn                                          | 4X12                | 1                    | 3                         | 265                                          | 600/1500                                                     | 300/1000                                                   | 10D/20D                                    |
| GYDTA-96Xn                                         | 8X12                | 2                    | 2                         | 265                                          |                                                              |                                                            |                                            |
| GYDTA-144Xn                                        | 12X12               | 3                    | 1                         | 265                                          |                                                              |                                                            |                                            |
| GYDTA-288Xn                                        | 24X12               | 4                    | 0                         | 295                                          |                                                              |                                                            |                                            |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# (GYXTW/GYXTZW)

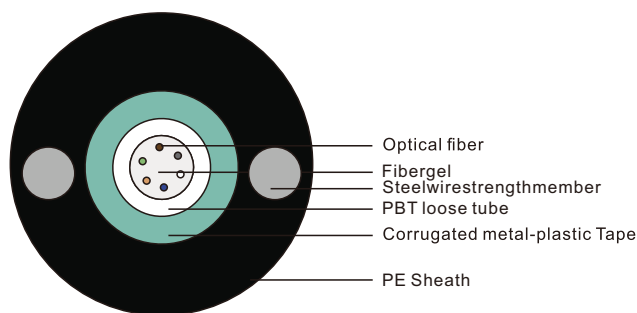
## Central tube type light armored optical cable



### Product Description

Please provide some specific examples of product dimensions and performance values.

Colored optical fibers are placed into loose tubes made of high-modulus, hydrolysis-resistant materials, and the loose tubes are filled with thixotropic water-blocking paste. The exterior of the loose tubes is longitudinally wrapped with a layer of double-sided coated steel tape, and the space between the steel tape and the loose tubes is filled with water-blocking cable paste to ensure the cable's compactness and longitudinal water resistance. Two parallel steel wires are placed on both sides after which a flame-retardant sheath is extruded to form the cable.



### Product Features

- It has excellent mechanical properties and temperature characteristics.
- The loose tube material features good hydrolysis resistance and excellent lateral pressure resistance.
- The tube is filled with special grease, providing critical protection for the optical fibers.
- The special sheath material has outstanding flame-retardant performance.
- The double-sided coated steel tape enhances the cable's moisture resistance.
- Two parallel steel wires ensure the stability and tensile performance of the optical cable.
- Small in diameter, light in weight, and easy to lay.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### ● Optical Characteristics

| Optical Cable Model<br>(Increasing by 2 Fibers)) | Number of Fibers | Reference Optical Cable Weight<br>(kg/km) | Allowable Tensile Force (Long-term /Short-term) (N) | Allowable Crush Force (Long-term/Short-term) (N) | Bending Radius (Static/Dynamic) (mm) |
|--------------------------------------------------|------------------|-------------------------------------------|-----------------------------------------------------|--------------------------------------------------|--------------------------------------|
| GYXTZW-24Xn                                      | 24               | 192                                       | 600/1500                                            | 300/1000                                         | 10D/20D                              |
| GYXTZW-24Xn                                      | 24               | 215                                       | 1000/3000                                           |                                                  |                                      |

Storage and service temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

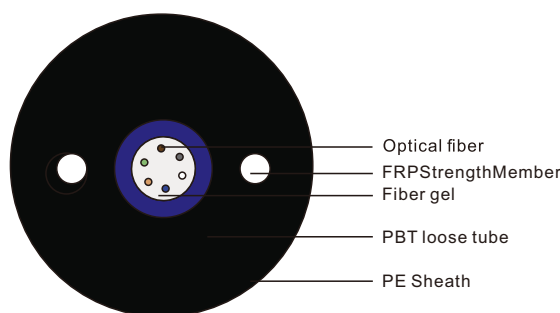
**(GYFXY)**

## Central tube type non-metallic non-armored optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus, hydrolysis-resistant materials, and the loose tubes are filled with thixotropic water-blocking paste. Water-blocking materials are applied outside the loose tubes to ensure the longitudinal water resistance of the optical cable. Two parallel glass fiber reinforced plastics (FRP) are placed on both sides, after which a polyethylene sheath is extruded to form the cable.



### Product Features

- ♦ It has good mechanical properties and temperature characteristics.
- ♦ The loose tube material has good hydrolysis resistance and excellent lateral pressure resistance.
- ♦ The tube is filled with special grease, providing critical protection for the optical fibers.
- ♦ The PE sheath has excellent UV radiation resistance.
- ♦ Two parallel FRPs ensure the stability and tensile performance of the optical cable.
- ♦ Small in diameter, light in weight, and easy to lay.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### ● Optical Characteristics

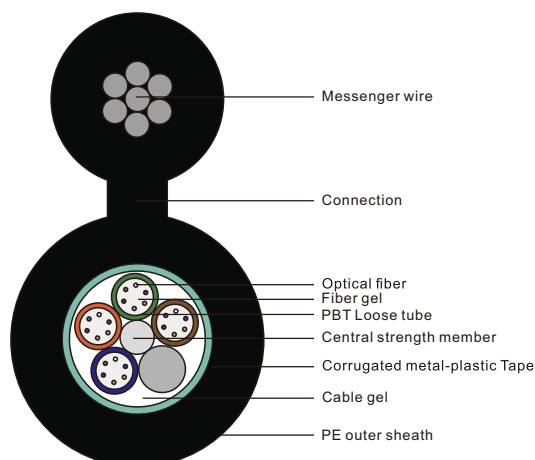
| Optical cable model<br>(in increments of 2 fibers) | Number of optical fibers | Reference cable weight<br>(kg/km) | Allowable tensile force<br>Long-term / Short-term<br>(N) | Allowable crushing force<br>Long-term / Short-term<br>(N) | Bending radius<br>Static / Dynamic (mm) |
|----------------------------------------------------|--------------------------|-----------------------------------|----------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------|
| GYFXY-24Xn                                         | 14~24                    | 96                                | 400/1500                                                 | 300/1000                                                  | 10D/20D                                 |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be adjusted according to the customer's specified requirements.

# (GY(F)TC8A GY(F)TC8Y GY(F)TC8S)

## Central tube type light armored optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus, hydrolysis-resistant material, and the loose tubes are filled with thixotropic water-blocking paste. At the center of the cable core is a metal strength member, around which the loose tubes (and filler ropes) are stranded to form a compact and circular cable core. The cable core is longitudinally wrapped with a layer of plastic-coated aluminum tape, and then integrated with steel wire strands into an 8-shaped polyethylene sheath.

### Product Features

- It has excellent mechanical properties and temperature characteristics.
- Steel strands used as self-supporting elements meet the tensile strength required for aerial optical cables, facilitating self-supporting aerial installation and reducing installation costs.
- The loose tube material has good hydrolysis resistance and excellent lateral pressure resistance.
- The tube is filled with special grease, providing critical protection for the optical fibers.
- The double-sided coated steel tape enhances the cable's moisture resistance.

### Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### Optical Characteristics

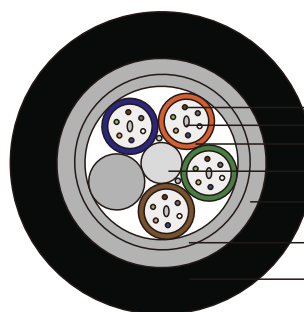
| Optical cable model<br>(in increments of 2 fibers) | Number of<br>optical fibers | Number of<br>sleeves | Number of filler<br>ropes | Reference cable weight<br>(kg/km) | Allowable tensile<br>force Long-term /<br>Short-term (N) | Allowable crushing<br>force Long-term /<br>Short-term (N) | Bending radius<br>Static / Dynamic<br>(mm) |
|----------------------------------------------------|-----------------------------|----------------------|---------------------------|-----------------------------------|----------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------|
| GYTC8A-24Xn                                        | 24                          | 4                    | 1                         | 158                               | According to<br>the diameter<br>of dropped wire          | 300/1000                                                  | 10D/20D                                    |
| GYTC8A-48Xn                                        | 48                          | 4                    | 1                         | 172                               |                                                          |                                                           |                                            |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

**(DCFA)**

# Central tube type non-metallic and non-armored optical cable



Optical fiber  
Water blocking Tape  
PP Loose tube  
Central strength member  
Corrugated metal-plastic Tape  
Water blocking Tape  
PE outer sheath

## Product Description

Colored optical fibers are placed into loose tubes made of high-modulus, hydrolysis-resistant material, and the loose tubes are filled with thixotropic water-blocking paste. At the center of the cable core is a non-metallic strength member (FRP), around which the loose tubes (and filler ropes) are stranded to form a compact and circular cable core. Both the loose tubes and the strength member are wrapped with water-blocking yarns. A water-blocking tape is longitudinally wrapped around the cable core, followed by a longitudinally wrapped plastic-coated aluminum tape, and finally a polyethylene sheath is extruded to form the cable.

## Product Features

- It has good mechanical properties and temperature characteristics.
- The loose tube material features good hydrolysis resistance and excellent lateral pressure resistance.
- Special water-blocking yarns are placed inside the loose tubes to provide critical protection for the optical fibers.
- The compact cable structure effectively prevents tube retraction.
- Plastic-coated aluminum tape moisture barrier.
- The PE sheath has excellent UV radiation resistance.

## ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

## ● Optical Characteristics

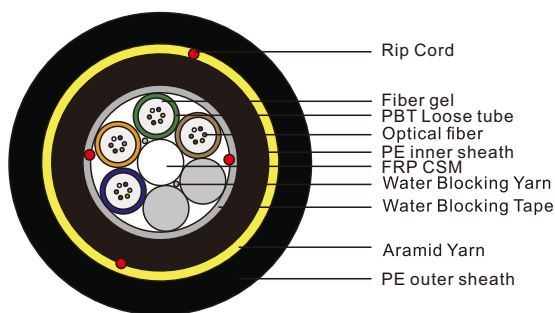
| Optical cable model<br>(in increments of 2 fibers) | Number of optical fibers | Number of sleeves | Number of filler ropes | Reference cable weight (kg/km) | Allowable tensile force Long-term / Short-term (N) | Allowable crushing force Long-term / Short-term (N) | Bending radius Static / Dynamic (mm) |
|----------------------------------------------------|--------------------------|-------------------|------------------------|--------------------------------|----------------------------------------------------|-----------------------------------------------------|--------------------------------------|
| DCFA-24Xn                                          | 24                       | 4                 | 2                      | 96                             | 400/1000                                           | 300/1000                                            | 10D/20D                              |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

# (ADSS)

## All-Dielectric Self-Supporting Optical Cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus hydrolysis-resistant material, and the loose tubes are filled with thixotropic water-blocking paste. At the center of the cable core is a metal strength member, around which the loose tubes (and filler ropes) are stranded to form a compact and circular cable core. The cable core, after being longitudinally wrapped with a layer of plastic-coated aluminum tape, is integrated with steel wire strands into an 8-shaped polyethylene sheath.

### Product Features

- Good mechanical properties and temperature characteristics
- Steel wire strands used as self-supporting components meet the tensile strength required for overhead optical cables, facilitating self-supporting overhead installation and reducing installation costs
- The loose tube material has excellent hydrolysis resistance and good lateral pressure resistance
- The tube is filled with special grease, providing key protection for optical fibers
- Double-sided plastic-coated steel tape enhances the moisture resistance of the cable

### Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850nm              | @1300nm   | @1310nm    | @1550nm    | @1550nm    | @1300nm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### Structural parameters

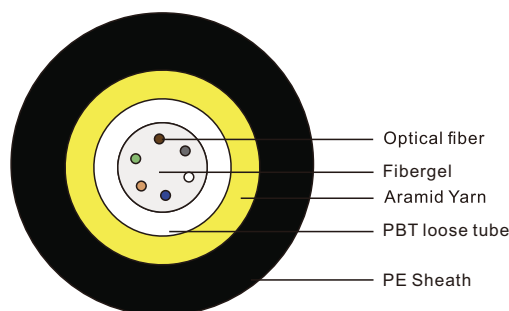
| Reference outer diameter (mm) | Reference weight (kg/km)          |     | Recommended daily maximum working tension (KN) | Maximum allowable working tension (KN) | Breaking strength (KN) | Tensile element cross-sectional area (mm <sup>2</sup> ) | Elastic modulus (kN/mm <sup>2</sup> ) | Thermal expansion coefficient ×10 <sup>-6</sup> /K | Suitable span (m)<br>Corresponding meteorological conditions (m) |     |     |     |
|-------------------------------|-----------------------------------|-----|------------------------------------------------|----------------------------------------|------------------------|---------------------------------------------------------|---------------------------------------|----------------------------------------------------|------------------------------------------------------------------|-----|-----|-----|
|                               | PE outer sheath / AT outer sheath |     |                                                |                                        |                        |                                                         |                                       |                                                    | A                                                                | B   | C   | D   |
| 11.8                          | 117                               | 124 | 1.5                                            | 4                                      | 10                     | 4.6                                                     | 7.6                                   | 1.8                                                | 160                                                              | 100 | 140 | 100 |
| 12.0                          | 121                               | 129 | 2.25                                           | 6                                      | 15                     | 7.6                                                     | 8.3                                   | 1.5                                                | 230                                                              | 150 | 200 | 150 |
| 12.3                          | 126                               | 134 | 3.0                                            | 8                                      | 20                     | 10.35                                                   | 9.45                                  | 1.3                                                | 300                                                              | 200 | 290 | 200 |
| 12.6                          | 133                               | 141 | 3.6                                            | 10                                     | 24                     | 13.8                                                    | 10.8                                  | 1.2                                                | 370                                                              | 250 | 350 | 250 |
| 12.8                          | 138                               | 145 | 4.5                                            | 12                                     | 30                     | 14.3                                                    | 11.8                                  | 1.0                                                | 420                                                              | 280 | 400 | 280 |
| 13.1                          | 145                               | 153 | 5.4                                            | 15                                     | 36                     | 18.4                                                    | 13.6                                  | 0.9                                                | 480                                                              | 320 | 460 | 320 |
| 13.5                          | 155                               | 163 | 6.75                                           | 18                                     | 45                     | 22.0                                                    | 16.4                                  | 0.6                                                | 570                                                              | 380 | 550 | 380 |
| 13.8                          | 163                               | 171 | 7.95                                           | 22                                     | 53                     | 26.4                                                    | 18.0                                  | 0.3                                                | 670                                                              | 320 | 650 | 460 |
| 14.4                          | 177                               | 186 | 9.0                                            | 26                                     | 60                     | 32.2                                                    | 19.1                                  | 0.1                                                | 750                                                              | 380 | 750 | 510 |
| 14.6                          | 182                               | 191 | 10.5                                           | 28                                     | 70                     | 33.0                                                    | 19.6                                  | 0.1                                                | 880                                                              | 460 | 800 | 560 |
| 14.8                          | 195                               | 204 | 12.75                                          | 34                                     | 85                     | 40.0                                                    | 20.1                                  | 0.1                                                | 880                                                              | 530 | 880 | 650 |

Storage and operating temperature: -40°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

(JET)

# Micro non-metallic air-blown optical cable



## Product Description

Colored optical fibers are placed into loose tubes made of high-modulus hydrolysis-resistant material, and the loose tubes are filled with non-toxic and harmless waterproof compounds. At the center of the cable core is a non-metallic strength member (FRP). The loose tubes (or filler ropes) are stranded around the central strength member to form a compact circular cable core. Water-blocking yarns are arranged outside the strength member, and then a black or colored high-density polyethylene sheath is extruded to form the cable.

## Product Features

- Non-metallic strength members are adopted, featuring small outer diameter, light weight, moderate hardness and softness, and the outer sheath has a very low friction coefficient, enabling a longer air-blown distance.
- The loose tube is filled with special grease, which provides buffer protection for optical fibers and achieves the waterproof performance of the cable.
- The compact cable structure effectively prevents tube retraction.

## Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.3db/km           | ≤1.2db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.5db/km           | ≤1.2db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

## Structural parameters

| Optical Cable Model<br>(in increments of 2 fibers) | Number of<br>optical fibers | Number of<br>loose<br>tubes | Number of<br>filler ropes | Cable<br>diameter | Reference Cable<br>Weight<br>(kg/km) | Allowable Tensile Force<br>Long-term / Short-<br>term<br>(N) | Allowable Crush Force<br>Long-term / Short-term<br>(N) | Bending Radius<br>Static / Dynamic<br>(mm) |
|----------------------------------------------------|-----------------------------|-----------------------------|---------------------------|-------------------|--------------------------------------|--------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------|
| GCYFTY-96Xn                                        | 96                          | 8                           | 0                         | 6.4               | 31                                   | 250/800                                                      | 150/450                                                | 10D/20D                                    |
| GCYFTY-144Xn                                       | 144                         | 12                          | 0                         | 8.3               | 49                                   | 250/800                                                      | 150/450                                                | 10D/20D                                    |

Storage and operating temperature: -30°C to +70°C.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

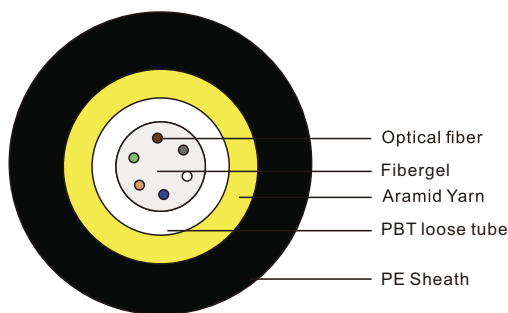
# (JET)

## Stranded non-metallic air-blown optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus hydrolysis-resistant material, and the loose tubes are filled with non-toxic and harmless waterproof compounds. At the center of the cable core is a non-metallic strength member (FRP). The loose tubes (or filler ropes) are stranded around the central strength member to form a compact circular cable core. Water-blocking yarns are arranged outside the strength member, and then a black or colored high-density polyethylene sheath is extruded to form the cable.



### Product Features

- Non-metallic strength members are adopted, featuring a small outer diameter, light weight, and moderate hardness. The outer sheath has an extremely low friction coefficient, allowing for a longer air-blown distance.
- The loose tube is filled with special grease, which provides buffer protection for the optical fibers and ensures the waterproof performance of the cable.
- The compact cable structure effectively prevents the retraction of the loose tube.

### ● Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.5db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

### ● Optical Characteristics

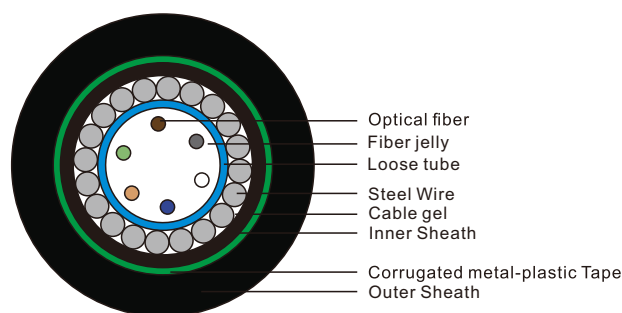
| Optical cable model<br>(in increments of 2 fibers) | Number of optical fibers | Number of loose tubes | Number of filler ropes | Cable diameter | Reference cable weight (kg/km) | Allowable Tensile Force Long-term / Short-term (N) | Allowable Crush Force Long-term / Short-term (N) | Bending Radius Static / Dynamic (mm) |
|----------------------------------------------------|--------------------------|-----------------------|------------------------|----------------|--------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------|
| GCYFTY-96Xn                                        | 96                       | 8                     | 0                      | 6.4            | 31                             | 250/800                                            | 150/450                                          | 10D/20D                              |
| GCYFTY-144Xn                                       | 144                      | 12                    | 0                      | 8.3            | 49                             | 250/800                                            | 150/450                                          | 10D/20D                              |

Storage and operating temperature: -30°C to +70°C.

Note: The dimensions and performance values of the product can be adjusted according to the specific requirements specified by the customer.

# (GYXTY53)

## Central tube type 53 - sheath anti - [specific hazard] optical cable



### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus material, which are filled with thixotropic water-blocking fiber paste. At the center of the cable core is a non-metallic strength member (FRP). The loose tubes (and filler ropes) are stranded around the central strength member to form a circular cable core, with the gaps in the cable core and the space between the cable core and the steel tape filled with water-blocking cable paste. A layer of polyethylene inner sheath is extruded over the cable core, followed by longitudinal wrapping with double-sided plastic-coated steel tape, and finally extrusion of a polyethylene outer sheath to complete the cable formation.

### Product Features

The loose tube material itself has excellent hydrolysis resistance and high strength.

- The tube is filled with special grease, providing critical sealing protection for optical fibers.
  - The PE sheath has excellent UV radiation resistance.
  - Single non-metallic central strength member.
  - Possesses excellent mechanical properties and temperature resistance.
  - The double-sided corrugated steel tape (PSP) enhances the cable's moisture resistance, and the corrugated part can better bond with PE, making the structure more robust.
- High-performance water-blocking materials prevent longitudinal water seepage in the cable.

### Structural Parameters

| Fiber Type | Attenuation (+20°C) |           |            |            | Bandwidth  |            | Numerical Aperture (NA) | Cable Cutoff Wavelength |
|------------|---------------------|-----------|------------|------------|------------|------------|-------------------------|-------------------------|
|            | @850mm              | @1300mm   | @1310mm    | @1550mm    | @1550mm    | @1300mm    |                         |                         |
| G.652      | -                   | -         | ≤0.36db/km | ≤0.22db/km | -          | -          | -                       | ≤1260nm                 |
| G.655      | -                   | -         | -          | ≤0.23db/km | -          | -          | -                       | ≤1450nm                 |
| 50/125μm   | ≤3.0db/km           | ≤1.0db/km | -          | -          | ≥500MHz·km | ≥500MHz·km | 0.200±0.015NA           | -                       |
| 62.5/125μm | ≤3.3db/km           | ≤1.0db/km | -          | -          | ≥200MHz·km | ≥500MHz·km | 0.275±0.015NS           | -                       |

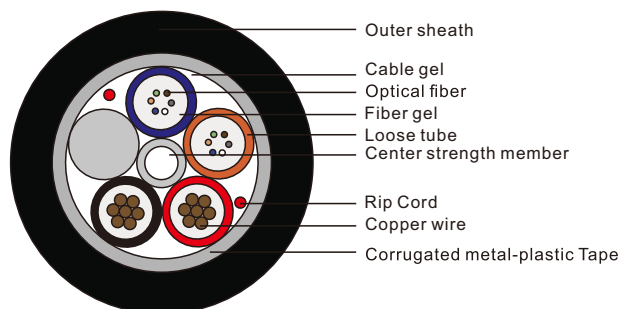
### Optical Characteristics

| Optical Cable Model<br>(Increasing by 2 fibers) | Number of optical fibers | Number of Sleeves | Number of Filler Ropes | Reference cable weight (kg/km) | Allowable tensile force<br>Long-term / Short-term (N) | Allowable crushing force<br>Long-term / Short-term (N) | Bending radius<br>Static / Dynamic (mm) |
|-------------------------------------------------|--------------------------|-------------------|------------------------|--------------------------------|-------------------------------------------------------|--------------------------------------------------------|-----------------------------------------|
| GYXTY53-24Xn                                    | 24                       | 4                 | 1                      | 96                             | 600/1500                                              | 300/1000                                               | 10D/20D                                 |
| GYXTY53-48Xn                                    | 48                       | 4                 | 1                      | 111                            |                                                       |                                                        |                                         |
| GYXTY53-96Xn                                    | 96                       | 8                 | 0                      | 168                            |                                                       |                                                        |                                         |
| GYXTY53-144Xn                                   | 144                      | 12                | 0                      | 228                            |                                                       |                                                        |                                         |
| GYXTY53-288Xn                                   | 288                      | 24                | 0                      | 283                            |                                                       |                                                        |                                         |

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.

# (GDTS/GDTA)

## Optoelectronic hybrid cable for access network



Outer sheath

Cable gel

Optical fiber

Fiber gel

Loose tube

Center strength member

Rip Cord

Copper wire

Corrugated metal-plastic Tape

### Product Description

Colored optical fibers are placed into loose tubes made of high-modulus, hydrolysis-resistant material, which are filled with thixotropic water-blocking paste. A layer of fine round steel wires is stranded around the loose tubes to form a compact, circular cable core, with water-blocking fillers packed into the gaps within the core. After longitudinally wrapping with plastic-coated aluminum tape, a polyethylene inner sheath is extruded. Then, after being armored with a single fine round steel wire, a polyethylene outer sheath is finally extruded to complete the cable.

### Product Features

- Boasts excellent mechanical properties and temperature resistance
- Double-layer single fine round steel wire armoring enhances the cable's tensile strength
- The loose tube material features superior hydrolysis resistance and strong lateral pressure resistance
- Special grease filled in the tubes offers critical protection for optical fibers
- Plastic-coated aluminum tape serves as a moisture-proof layer
- Compact in size, lightweight, and easy to install

### Electrical properties of insulated conductors

| Optical Cable Model<br>(Increasing by 2 fibers) | Reference<br>Outer Diameter<br>(mm) | Reference Weight<br>(kg/km) | Allowable Tensile<br>Force Long-term / Short-term (N) | Allowable Crushing<br>Force Long-term / Short-term (N) | Reference<br>Structure |
|-------------------------------------------------|-------------------------------------|-----------------------------|-------------------------------------------------------|--------------------------------------------------------|------------------------|
| GDTS-02~24Xn+2X1.5                              | 11.6                                | 157                         | 600/1500                                              | 300/1000                                               | I型结构                   |
| GDTS-02~24Xn+2X2.5                              | 12.5                                | 190                         | 600/1500                                              | 300/1000                                               | II型结构                  |
| GDTS-02~24Xn+2X4.0                              | 13.6                                | 241                         | 600/1500                                              | 300/1000                                               | II型结构                  |
| GDTS-02~24Xn+2X5.0                              | 15.0                                | 282                         | 600/1500                                              | 300/1000                                               | III型结构                 |
| GDTS-02~24Xn+2X6.0                              | 15.7                                | 300                         | 600/1500                                              | 300/1000                                               | III型结构                 |
| GDTS-02~24Xn+2X8.0                              | 16.9                                | 383                         | 600/1500                                              | 300/1000                                               | III型结构                 |

#### Note: Xn represents the optical fiber type.

2X1.5/2X2.5/2X4.0/2X6.0/2X8.0 indicates the quantity and specification of conductors.

Optoelectronic hybrid cables with different conductor specifications and quantities can be produced according to user requirements.

Optoelectronic composite cables with different optical fiber core counts can be manufactured as per customer requests.

### Electrical properties of insulated conductors

| Cross-sectional<br>area<br>of conductor<br>(mm <sup>2</sup> ) | Maximum DC resistance<br>of a single conductor<br>(20°C) (Ω/km) | Insulation resistance (20°C) (MΩ.km)                                        | Insulation voltage strength KV, DC 1minute |                                 |                                  |
|---------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------|---------------------------------|----------------------------------|
|                                                               |                                                                 | Between each insulated conductor and other metal connected within the cable | Between Conductors                         | Conductors and Metal Armor Tape | Between conductor and steel wire |
| 1.5                                                           | 13.3                                                            | Not less than 10000                                                         | 5                                          | 5                               | 3                                |
| 2.5                                                           | 7.98                                                            |                                                                             |                                            |                                 |                                  |
| 4.0                                                           | 4.95                                                            |                                                                             |                                            |                                 |                                  |
| 5.0                                                           | 3.88                                                            |                                                                             |                                            |                                 |                                  |
| 6.0                                                           | 3.30                                                            |                                                                             |                                            |                                 |                                  |
| 8.0                                                           | 2.47                                                            |                                                                             |                                            |                                 |                                  |

Note: The relevant performance of feeder conductors complies with GB/T 3956-2008. The electrical performance of feeders meets the requirements of GB/T 5023.3-2007.

Note: The dimensions and performance values of the product can be customized according to customer-specific requirements.



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